TEXORA · COURSE NOTES · LECTURE SHEET

Optics

The behaviour of light — reflection and refraction, total internal reflection, lenses and prisms, interference, diffraction and polarisation, with the instruments of the Physics-I laboratory.

PHY-1103 · PHYSICS-I · TOPIC 6 · LEVEL 1 TERM 1

1. Reflection, refraction and refractive index

Light travels in straight lines (rays) until it meets a boundary. **Reflection** obeys two laws: the angle of reflection equals the angle of incidence, and the incident ray, reflected ray and normal lie in one plane. At a boundary into a different medium the ray bends — **refraction** — following **Snell's law**:

$$\sin i / \sin r = \mu \text{ (a constant for the pair of media)}$$

μ, the **refractive index**, is also the ratio of light's speed in vacuum (or air) to its speed in the medium: μ = c/v. The ray bends *toward* the normal entering a denser (higher-μ) medium and *away* entering a rarer one. Everyday bookkeeping follows: a pool looks shallower than it is (real/apparent depth = μ), a straw bends at the water line, and the sessional course measures μ of glass with a spectrometer prism and of liquids with a lens and mirror.

2. Total internal reflection and the critical angle

Going from denser to rarer, the refracted ray bends ever further from the normal as i grows, until at the **critical angle** C it skims along the boundary (r = 90°), with sin C = 1/μ. Beyond C there is no refracted ray at all — the light is **totally internally reflected**, a perfect mirror with no silvering. For glass (μ = 1.5), C ≈ 41.8°; for water (μ = 1.33), C ≈ 48.8°. The applications built an era: **optical fibres** — hair-thin glass strands that pipe light around any bend by repeated total internal reflection (telecommunications, endoscopes, and the optical-fibre sensors of technical textiles), the sparkle of a cut diamond (μ ≈ 2.4, C ≈ 24°, so light is trapped and returned), and reflecting prisms in binoculars and periscopes.

3. Lenses: formula, magnification, power

A lens forms images by refraction at two curved surfaces. With the Cartesian sign convention (distances measured from the lens, positive in the direction of the incident light's travel), the **lens formula** is

$$1/f = 1/v - 1/u$$

where f is the focal length, u the object distance and v the image distance (u enters negative for a real object on the incoming side under this convention — follow your laboratory manual's convention consistently, since sign conventions differ between texts and the physics does not). **Magnification** m = v/u (image height/object height); the **power** of a lens is P = 1/f in metres, measured in dioptres (D) — + for converging, − for diverging. The lens-maker's formula, $1/f = (\mu - 1)(1/R_1 - 1/R_2)$, shows where f comes from: the glass's μ and the two surface curvatures.

> **WORKED EXAMPLE — A CONVERGING LENS**
>
> A convex lens of f = +20 cm images an object placed 30 cm before it. In the "real is positive" laboratory convention the formula reads 1/f = 1/v + 1/u, so 1/v = 1/f − 1/u = 1/20 − 1/30 = (3 − 2)/60 = 1/60, giving v = **60 cm** beyond the lens — a real, inverted image with magnification m = v/u = 2, twice life size. (The two printed forms of the lens formula differ only in sign convention; pick the one your manual uses and never mix them mid-calculation.)

Combinations matter practically: two thin lenses in contact add powers ($P = P_1 + P_2$), which is how spectacles correct eyes and how the microscope — the fibre laboratory's chief instrument — stacks an objective and an eyepiece.

4. The prism and dispersion

A ray through a prism bends toward the base at both faces; the total **deviation** δ depends on the angle of incidence and passes through a **minimum** δ_m at symmetric passage, where the prism formula gives μ exactly:

$$\mu = \sin\left[\,(A + \delta_m)/2\,\right] / \sin(A/2)$$

with A the prism angle — the spectrometer measurement of the sessional course. Because μ differs slightly with wavelength (greater for violet than for red), a prism also **disperses** white light into a spectrum: violet deviates most, red least. Dispersion is the working principle of the spectroscope, and in reverse — Newton's recombination — the proof that white light is a mixture.

5. Interference: Young's double slit

Two coherent sources — in practice, two slits lit by one source — produce a pattern of bright and dark **fringes** where waves arrive in step or out of step. The fringe width (bright-to-bright spacing) is

$$\beta = \lambda D / d$$

(D the slit-to-screen distance, d the slit separation). Worked: $\lambda = 600$ nm, $D = 1.0$ m, $d = 0.50$ mm gives $\beta = 600 \times 10^{-9} \times 1.0 / 5.0 \times 10^{-4} = 1.2 \times 10^{-3}$ m = **1.2 mm** — visible fringes from an invisible wavelength, which is the point: the pattern is a ruler for λ. Thin films show the same physics in colour (oil on water, soap bubbles, Newton's rings — the sessional's own arrangement for measuring a lens surface's radius of curvature from the rings' diameters).

6. Diffraction and the grating

Light also bends around edges and through apertures — **diffraction** — dramatically when the opening is comparable to λ. A **diffraction grating** (hundreds of lines per millimetre) turns this into the sharpest wavelength instrument of the course: maxima appear where

$$d \sin \theta = n \lambda$$

(d the grating spacing = 1/(lines per unit length), n the order). Worked: a grating of 600 lines/mm has $d = 1/(600 \times 10^3)$ m $= 1.667 \times 10^{-6}$ m; for $\lambda = 600$ nm in first order, $\sin \theta = 600 \times 10^{-9} / 1.667 \times 10^{-6} = 0.36$, $\theta \approx$ **21.1°** — the sessional's discharge-tube measurement. Single-slit diffraction and the resolving power of instruments (why microscopes cannot see below ~$\lambda/2$) complete the standard picture.

7. Polarisation

Light is a transverse wave; in ordinary light the electric vibrations point in all directions perpendicular to travel. **Polarisation** confines the vibration to one plane. It is produced by selective absorption (Polaroid sheets — stretched polymer with aligned molecules, another textile-relative), by reflection (at **Brewster's angle**, $\tan i_p = \mu$, the reflected light is completely polarised), by double refraction in crystals like calcite, and by scattering (skylight). The polariscope and polarising microscope built on it identify stressed transparent materials — and, in the fibre laboratory, distinguish fibres by their **birefringence**: under crossed polars each fibre family shows characteristic interference colours, a standard identification test in textile analysis.

8. The textile thread

Optics is the fibre laboratory's toolkit. The **polarising microscope** identifies fibres by birefringence; the **spectrophotometer** measures colour by the prism/grating physics of §§4–6 married to the Lambert–Beer law of Chemistry-I; **lustre** — the fibre's fashion value — is controlled reflection from its surface and cross-section; optical brighteners work by fluorescence (Chemistry-I, Topic 9); and optical fibres themselves have entered technical textiles as sensors woven into composites and garments. Even the light booth where shades are matched is applied optics: standard illuminants, controlled geometry, no surprises.

9. Glossary — key terms of this topic

TERM	MEANING
Refractive index (μ)	sin i / sin r for a medium pair; also c/v, the slowing factor of light in the medium.
Critical angle	The incidence angle in the denser medium for which the refracted ray grazes the boundary: sin C = 1/μ.
Total internal reflection	Complete reflection at a denser → rarer boundary beyond the critical angle.
Power of a lens	P = 1/f (f in metres), in dioptres; positive for converging lenses.
Dispersion	The splitting of white light into a spectrum because μ varies with wavelength.
Fringe width (β)	Spacing of interference fringes in Young's experiment: β = λD/d.
Grating equation	d sin θ = nλ gives the directions of a grating's principal maxima.
Birefringence	Double refraction — two refractive indices for two vibration directions in an anisotropic material; a fibre-identification property.

10. Key points and summary

- Snell's law and μ = c/v govern refraction; beyond the critical angle (sin C = 1/μ) total internal reflection carries optical fibres and prism instruments.
- Lenses obey 1/f = 1/v − 1/u (sign convention of your manual), with magnification v/u and power 1/f in dioptres; the prism at minimum deviation yields μ exactly, and μ(λ) gives dispersion.
- Wave optics completes the picture: interference fringes β = λD/d, grating maxima d sin θ = nλ, and polarisation — Brewster's angle, Polaroids, and the birefringence by which the fibre laboratory names its fibres.

11. Practice

Answer in your own words or with full working; brief model answers follow.

1. A ray enters glass (μ = 1.5) from air at i = 45°. Find the angle of refraction.
2. Find the critical angle for water (μ = 1.33) and say why a diver looking up sees the whole sky compressed into a cone.
3. A lens has power +2.5 D. Find its focal length and state whether it converges or diverges light.
4. A prism of angle 60° shows minimum deviation 37° for a ray. Find μ of the prism material.

5. In a double-slit experiment with d = 0.25 mm and D = 0.80 m, fringes are 1.6 mm apart. Find the wavelength of the light.
6. A grating has 500 lines/mm. Find the angle of the second-order maximum for λ = 500 nm.
7. State Brewster’s law and compute the polarising angle for glass of μ = 1.5.
8. Two fibres look alike under the ordinary microscope but show different interference colours between crossed polars. Name the property responsible and the instrument used.

Answer key

1. $\sin r = \sin 45°/1.5 = 0.7071/1.5 = 0.4714 \rightarrow r \approx 28.1°$.
2. $\sin C = 1/1.33 = 0.752 \rightarrow C \approx 48.8°$. Rays from the sky strike the surface at all angles up to 90°; refraction compresses them into angles below C inside the water, so the whole 180° sky fits within a cone of half-angle C (the rest of the surface view is total internal reflection of the underwater scene).
3. f = 1/P = 1/2.5 = 0.40 m = 40 cm; positive power — a converging (convex) lens.
4. $\mu = \sin[(60 + 37)/2] / \sin(30) = \sin 48.5° / 0.5 = 0.749/0.5 \approx 1.50$.
5. $\lambda = \beta\, d / D = (1.6 \times 10^{-3} \times 0.25 \times 10^{-3}) / 0.80 = 4.0 \times 10^{-7} / 0.80 = 5.0 \times 10^{-7}$ m = 500 nm.
6. $d = 1/(500 \times 10^{3}) = 2.0 \times 10^{-6}$ m; $\sin\theta = n\lambda/d = 2 \times 500 \times 10^{-9} / 2.0 \times 10^{-6} = 0.50 \rightarrow \theta = 30°$.
7. $\tan i_p = \mu$; $i_p = \tan^{-1}(1.5) \approx 56.3°$. At this incidence the reflected ray is completely plane-polarised.
8. Birefringence (double refraction, differing μ for the two vibration directions), examined with a polarising microscope.

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